

I. Introduction:

Research Design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure.

Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design.

It constitutes the blueprint for the collection, measurement and analysis of data. As such the design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data.

Designing decisions happen to be in respect of:

- (i) What is the study about?
- (ii) Why is the study being made?
- (iii) Where will the study be carried out?
- (iv) What type of data is required?
- (v) Where can the required data be found?
- (vi) What periods of time will the study include?
- (vii) What will be the sample design?
- (viii) What techniques of data collection will be used?
- (ix) How will the data be analysed?
- (x) In what style will the report be prepared?

Keeping in view the above stated design decisions, one may split the overall research design into the following **parts: (Parts of Research Design):**

- (a) the **sampling design** which deals with the method of selecting items to be observed for the given study.
- (b) the **observational design** which relates to the conditions under which the observations are to be made.
- (c) the **statistical design** which concerns the question of how many items are to be observed and how the information and data gathered are to be analysed.
- (d) the **operational design** which deals with the techniques by which the procedures specified in the sampling, statistical and observational designs can be carried out.

II. FEATURES OF A GOOD DESIGN:

A good design is often characterised by adjectives like flexible, appropriate, efficient, economical and so on. Generally, the design which minimises bias and maximises the reliability of the data collected and analysed is considered a good design. The design which gives the smallest experimental error is supposed to be the best design in many investigations. Similarly, a design which yields maximal information and provides an opportunity for considering many different aspects of a problem is considered most appropriate and efficient design in respect of many research problems.

- the means of obtaining information;
- the availability and skills of the researcher and his staff, if any;
- the objective of the problem to be studied;
- the nature of the problem to be studied; and

- the availability of time and money for the research work.

III. NEED FOR RESEARCH DESIGN:

Research design is needed because it facilitates the smooth sailing of the various research operations, thereby making research as efficient as possible yielding maximal information with minimal expenditure of effort, time and money. Just as for better, economical and attractive construction of a house, we need a blueprint (or what is commonly called the map of the house) well thought out and prepared by an expert architect, similarly we need a research design or a plan in advance of data collection and analysis for our research project. Research design stands for advance planning of the methods to be adopted for collecting the relevant data and the techniques to be used in their analysis, keeping in view the objective of the research and the availability of staff, time and money. Preparation of the research design should be done with great care as any error in it may upset the entire project. Research design, in fact, has a great bearing on the reliability of the results arrived at and as such constitutes the firm foundation of the entire edifice of the research work.

IV. IMPORTANT CONCEPTS RELATING TO RESEARCH DESIGN

- Dependent and independent variables
- Extraneous variable
- Control
- Confounded relationship
- Research hypothesis
- Experimental and non-experimental hypothesis-testing research
- Experimental and control groups
- Treatments
- Experiment
- Experimental unit(s)

1. Dependent and independent variables

A concept which can take on different quantitative values is called a variable.

- Example: weight, height, income, etc., are all variables
- Continuous variable – value with decimal point (Ex. Height)
- Discrete variable – integer value. (Ex. Age)
- Height depends on Age. **height is a dependent variable**, Age is an independent variable.

2. Extraneous variable

Independent variables that are not related to the purpose of the study, but may affect the dependent variable are termed as extraneous variables.

Example: there is a relationship between children's achievement in Maths and special training.

In this special training is an independent variable and achievement in Maths is a dependent variable.

Intelligence may as well affect the achievement in Maths, but since it is not related to the purpose of the study undertaken by the researcher, it will be termed as an **extraneous variable (Intelligence is extraneous variable)**.

3. Control

In experimental researches, the term 'control' is used to refer to restrain experimental conditions. One important characteristic of a good research design is to minimize the influence or effect of extraneous variables.

4. Confounded relationship

When the dependent variable is not free from the influence of extraneous variables, the relationship between the dependent and independent variables is said to be confounded (mixed) by an extraneous variable.

5. Research hypothesis

When a prediction or a hypothesised relationship is to be tested by scientific methods, it is termed as research hypothesis.

6. Experimental and non-experimental hypothesis-testing research

When the purpose of research is to test a research hypothesis, it is termed as hypothesis-testing research. It can be of the experimental design or of the non-experimental design. Research in which the independent variable is manipulated is termed 'experimental hypothesis-testing research' and research in which an independent variable is not manipulated is called 'non-experimental hypothesis-testing research'. For instance, suppose a researcher wants to study whether intelligence affects reading ability for a group of students and for this purpose he randomly selects 50 students and tests their intelligence and reading ability by calculating the coefficient of correlation between the two sets of scores. This is an example of non-experimental hypothesis-testing research because herein the independent variable, intelligence, is not manipulated. But now suppose that our researcher randomly selects 50 students from a group of students who are to take a course in statistics and then divides them into two groups by randomly assigning 25 to Group A, the usual studies programme, and 25 to Group B, the special studies programme. At the end of the course, he administers a test to each group in order to judge the effectiveness of the training programme on the student's performance-level. This is an example of experimental hypothesis-testing research because in this case the independent variable, namely, the type of training programme, is manipulated.

7. Experimental and control groups

When a group is exposed to usual conditions, it is termed a 'control group', but when the group is exposed to some novel or special condition, it is termed an 'experimental group'.

Example: a) Special training given for students is termed as experimental group'

b) Usual studies program (No Special training given) is termed as control group.

8. Treatments

The different conditions under which experimental and control groups are put are usually referred to as 'treatments'.

Example: Special training and usual studies program are different conditions

9. Experiment

The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an experiment.

10. Experimental unit(s)

The pre-determined plots or the blocks, where different treatments are used, are known as experimental units.

V. Basic principles of experimental designs:

a) The Principle of Replication:

According to the Principle of Replication, the experiment should be repeated more than once.

Thus, each treatment is applied in many experimental units instead of one. By doing so the statistical accuracy of the experiments is increased.

For example, suppose we are to examine the effect of two varieties of rice. For this purpose we may divide the field into two parts and grow one variety in one part and the other variety in the other part. We can then compare the yield of the two parts and draw conclusion on that basis.

But if we are to apply the **principle of replication to this experiment**, then we first divide the field into several parts, grow one variety in half of these parts and the other variety in the remaining parts. We can then collect the data of yield of the two varieties and draw conclusion by comparing the same. The result so obtained will be more reliable in comparison to the conclusion we draw without applying the principle of replication. The entire experiment can even be repeated several times for better results.

b) The Principle of Randomization:

This principle indicates that we should design or plan the experiment in such a way that the variations caused by extraneous factors can all be combined under the general heading of “chance”.

For Example, if we grow one variety of rice, say, in the first half of the parts of a field and the other variety is grown in the other half, then it is just possible that the soil fertility may be different in the first half in comparison to the other half. If this is so, our results would not be realistic. In such a situation, we may assign the variety of rice to be grown in different parts of the field on the basis of some random sampling technique i.e., we may apply randomization principle and protect ourselves against the effects of the extraneous factors (soil fertility differences in the given case). As such, through the application of the principle of randomization, we can have a better estimate of the experimental error.

c) The Principle of Local Control:

The extraneous factor, known as source of variability, is made to vary deliberately over a wide range and this needs to be done in such a way that the variability can be measured and hence eliminated from the experimental error.

For example, varieties of rice in our case, the extraneous factor (soil fertility in our case). we first divide the field into several homogeneous parts, known as blocks, and then each such block is divided into parts equal to the number of treatments. Then the treatments are randomly assigned to these parts of a block.

Basic Principles of Experimental Design:

